

STATISTICAL TABLES

统计分布表

数学与统计学学科组制

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1 Standard normal distribution

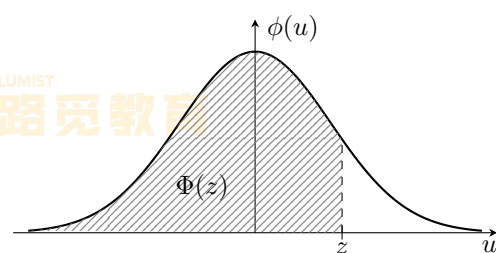
The table below contains the probability

$$\Phi(z) = \Pr(Z \leq z) = \int_{-\infty}^z \phi(u) \, du = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) \, du$$

for a normally distributed random variable Z with expected value 0 and variance 1, i.e. $Z \sim \mathcal{N}(0, 1)$.

The table only contains the probabilities for $z \geq 0$. For $z < 0$, the probabilities can be obtained using the identity

$$\Phi(z) = 1 - \Phi(-z).$$



z	z									
	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

2 Student's t distribution

The table below contains the quantiles of Student's t distribution with ν degrees of freedom. For $0 < p < 1$ the quantile is the value of x for which

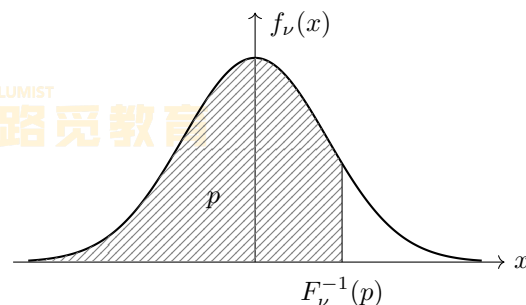
$$\Pr(X \leq x) = p, \quad \text{where } X \sim t(\nu).$$

Thus

$$x = F_{\nu}^{-1}(p).$$

The table only contains the quantiles for $p \geq \frac{1}{2}$. For $p < \frac{1}{2}$ quantiles can be obtained by exploiting the symmetry of the t distribution:

$$F_{\nu}^{-1}(p) = -F_{\nu}^{-1}(1 - p).$$



df (ν)	p											
	0.6	0.7	0.75	0.8	0.85	0.9	0.95	0.975	0.99	0.995	0.999	0.9995
1	0.3249	0.7265	1.0000	1.3764	1.9626	3.0777	6.3138	12.706	31.821	63.657	318.31	636.62
2	0.2887	0.6172	0.8165	1.0607	1.3862	1.8856	2.9200	4.3027	6.9646	9.9248	22.327	31.599
3	0.2767	0.5844	0.7649	0.9785	1.2498	1.6377	2.3534	3.1824	4.5407	5.8409	10.215	12.924
4	0.2707	0.5686	0.7407	0.9410	1.1896	1.5332	2.1318	2.7764	3.7469	4.6041	7.1732	8.6103
5	0.2672	0.5594	0.7267	0.9195	1.1558	1.4759	2.0150	2.5706	3.3649	4.0321	5.8934	6.8688
6	0.2648	0.5534	0.7176	0.9057	1.1342	1.4398	1.9432	2.4469	3.1427	3.7074	5.2076	5.9588
7	0.2632	0.5491	0.7111	0.8960	1.1192	1.4149	1.8946	2.3646	2.9980	3.4995	4.7853	5.4079
8	0.2619	0.5459	0.7064	0.8889	1.1081	1.3968	1.8595	2.3060	2.8965	3.3554	4.5008	5.0413
9	0.2610	0.5435	0.7027	0.8834	1.0997	1.3830	1.8331	2.2622	2.8214	3.2498	4.2968	4.7809
10	0.2602	0.5415	0.6998	0.8791	1.0931	1.3722	1.8125	2.2281	2.7638	3.1693	4.1437	4.5869
11	0.2596	0.5399	0.6974	0.8755	1.0877	1.3634	1.7959	2.2010	2.7181	3.1058	4.0247	4.4370
12	0.2590	0.5386	0.6955	0.8726	1.0832	1.3562	1.7823	2.1788	2.6810	3.0545	3.9296	4.3178
13	0.2586	0.5375	0.6938	0.8702	1.0795	1.3502	1.7709	2.1604	2.6503	3.0123	3.8520	4.2208
14	0.2582	0.5366	0.6924	0.8681	1.0763	1.3450	1.7613	2.1448	2.6245	2.9768	3.7874	4.1405
15	0.2579	0.5357	0.6912	0.8662	1.0735	1.3406	1.7531	2.1314	2.6025	2.9467	3.7328	4.0728
16	0.2576	0.5350	0.6901	0.8647	1.0711	1.3368	1.7459	2.1199	2.5835	2.9208	3.6862	4.0150
17	0.2573	0.5344	0.6892	0.8633	1.0690	1.3334	1.7396	2.1098	2.5669	2.8982	3.6458	3.9651
18	0.2571	0.5338	0.6884	0.8620	1.0672	1.3304	1.7341	2.1009	2.5524	2.8784	3.6105	3.9216
19	0.2569	0.5333	0.6876	0.8610	1.0655	1.3277	1.7291	2.0930	2.5395	2.8609	3.5794	3.8834
20	0.2567	0.5329	0.6870	0.8600	1.0640	1.3253	1.7247	2.0860	2.5280	2.8453	3.5518	3.8495
21	0.2566	0.5325	0.6864	0.8591	1.0627	1.3232	1.7207	2.0796	2.5176	2.8314	3.5272	3.8193
22	0.2564	0.5321	0.6858	0.8583	1.0614	1.3212	1.7171	2.0739	2.5083	2.8188	3.5050	3.7921
23	0.2563	0.5317	0.6853	0.8575	1.0603	1.3195	1.7139	2.0687	2.4999	2.8073	3.4850	3.7676
24	0.2562	0.5314	0.6848	0.8569	1.0593	1.3178	1.7109	2.0639	2.4922	2.7969	3.4668	3.7454
25	0.2561	0.5312	0.6844	0.8562	1.0584	1.3163	1.7081	2.0595	2.4851	2.7874	3.4502	3.7251
26	0.2560	0.5309	0.6840	0.8557	1.0575	1.3150	1.7056	2.0555	2.4786	2.7787	3.4350	3.7066
27	0.2559	0.5306	0.6837	0.8551	1.0567	1.3137	1.7033	2.0518	2.4727	2.7707	3.4210	3.6896
28	0.2558	0.5304	0.6834	0.8546	1.0560	1.3125	1.7011	2.0484	2.4671	2.7633	3.4082	3.6739
29	0.2557	0.5302	0.6830	0.8542	1.0553	1.3114	1.6991	2.0452	2.4620	2.7564	3.3962	3.6594
30	0.2556	0.5300	0.6828	0.8538	1.0547	1.3104	1.6973	2.0423	2.4573	2.7500	3.3852	3.6460
40	0.2550	0.5286	0.6807	0.8507	1.0500	1.3031	1.6839	2.0211	2.4233	2.7045	3.3069	3.5510
50	0.2547	0.5278	0.6794	0.8489	1.0473	1.2987	1.6759	2.0086	2.4033	2.6778	3.2614	3.4960
60	0.2545	0.5272	0.6786	0.8477	1.0455	1.2958	1.6706	2.0003	2.3901	2.6603	3.2317	3.4602
80	0.2542	0.5265	0.6776	0.8461	1.0432	1.2922	1.6641	1.9901	2.3739	2.6387	3.1953	3.4163
100	0.2540	0.5261	0.6770	0.8452	1.0418	1.2901	1.6602	1.9840	2.3642	2.6259	3.1737	3.3905
120	0.2539	0.5258	0.6765	0.8446	1.0409	1.2886	1.6577	1.9799	2.3578	2.6174	3.1595	3.3735
150	0.2538	0.5255	0.6761	0.8440	1.0400	1.2872	1.6551	1.9759	2.3515	2.6090	3.1455	3.3566
200	0.2537	0.5252	0.6757	0.8434	1.0391	1.2858	1.6525	1.9719	2.3451	2.6006	3.1315	3.3398
500	0.2535	0.5247	0.6750	0.8423	1.0375	1.2832	1.6479	1.9647	2.3338	2.5857	3.1066	3.3101

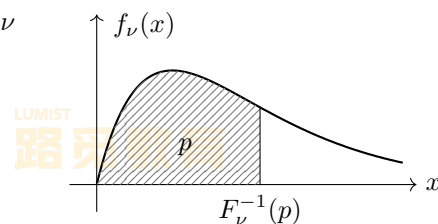
3 χ^2 distribution

The table below contains the quantiles of the χ^2 (chi-squared) distribution with ν degrees of freedom. For $0 < p < 1$ the quantile is the value of x for which

$$\Pr(X \leq x) = p, \quad \text{where } X \sim \chi^2(\nu).$$

Thus

$$x = F_{\nu}^{-1}(p).$$



df (ν)	p											
	0.005	0.01	0.025	0.05	0.1	0.5	0.9	0.95	0.975	0.99	0.995	0.999
1	0.0000	0.0002	0.0010	0.0039	0.0158	0.4549	2.7055	3.8415	5.0239	6.6349	7.8794	10.828
2	0.0100	0.0201	0.0506	0.1026	0.2107	1.3863	4.6052	5.9915	7.3778	9.2103	10.597	13.816
3	0.0717	0.1148	0.2158	0.3518	0.5844	2.3660	6.2514	7.8147	9.3484	11.345	12.838	16.266
4	0.2070	0.2971	0.4844	0.7107	1.0636	3.3567	7.7794	9.4877	11.143	13.277	14.860	18.467
5	0.4117	0.5543	0.8312	1.1455	1.6103	4.3515	9.2364	11.070	12.833	15.086	16.750	20.515
6	0.6757	0.8721	1.2373	1.6354	2.2041	5.3481	10.645	12.592	14.449	16.812	18.548	22.458
7	0.9893	1.2390	1.6899	2.1673	2.8331	6.3458	12.017	14.067	16.013	18.475	20.278	24.322
8	1.3444	1.6465	2.1797	2.7326	3.4895	7.3441	13.362	15.507	17.535	20.090	21.955	26.124
9	1.7349	2.0879	2.7004	3.3251	4.1682	8.3428	14.684	16.919	19.023	21.666	23.589	27.877
10	2.1559	2.5582	3.2470	3.9403	4.8652	9.3418	15.987	18.307	20.483	23.209	25.188	29.588
11	2.6032	3.0535	3.8157	4.5748	5.5778	10.341	17.275	19.675	21.920	24.725	26.757	31.264
12	3.0738	3.5706	4.4038	5.2260	6.3038	11.340	18.549	21.026	23.337	26.217	28.300	32.909
13	3.5650	4.1069	5.0088	5.8919	7.0415	12.340	19.812	22.362	24.736	27.688	29.819	34.528
14	4.0747	4.6604	5.6287	6.5706	7.7895	13.339	21.064	23.685	26.119	29.141	31.319	36.123
15	4.6009	5.2293	6.2621	7.2609	8.5468	14.339	22.307	24.996	27.488	30.578	32.801	37.697
16	5.1422	5.8122	6.9077	7.9616	9.3122	15.338	23.542	26.296	28.845	32.000	34.267	39.252
17	5.6972	6.4078	7.5642	8.6718	10.085	16.338	24.769	27.587	30.191	33.409	35.718	40.790
18	6.2648	7.0149	8.2307	9.3905	10.865	17.338	25.989	28.869	31.526	34.805	37.156	42.312
19	6.8440	7.6327	8.9065	10.117	11.651	18.338	27.204	30.144	32.852	36.191	38.582	43.820
20	7.4338	8.2604	9.5908	10.851	12.443	19.337	28.412	31.410	34.170	37.566	39.997	45.315
21	8.0337	8.8972	10.283	11.591	13.240	20.337	29.615	32.671	35.479	38.932	41.401	46.797
22	8.6427	9.5425	10.982	12.338	14.041	21.337	30.813	33.924	36.781	40.289	42.796	48.268
23	9.2604	10.196	11.689	13.091	14.848	22.337	32.007	35.172	38.076	41.638	44.181	49.728
24	9.8862	10.856	12.401	13.848	15.659	23.337	33.196	36.415	39.364	42.980	45.559	51.179
25	10.520	11.524	13.120	14.611	16.473	24.337	34.382	37.652	40.646	44.314	46.928	52.620
26	11.160	12.198	13.844	15.379	17.292	25.336	35.563	38.885	41.923	45.642	48.290	54.052
27	11.808	12.879	14.573	16.151	18.114	26.336	36.741	40.113	43.195	46.963	49.645	55.476
28	12.461	13.565	15.308	16.928	18.939	27.336	37.916	41.337	44.461	48.278	50.993	56.892
29	13.121	14.256	16.047	17.708	19.768	28.336	39.087	42.557	45.722	49.588	52.336	58.301
30	13.787	14.953	16.791	18.493	20.599	29.336	40.256	43.773	46.979	50.892	53.672	59.703
40	20.707	22.164	24.433	26.509	29.051	39.335	51.805	55.758	59.342	63.691	66.766	73.402
50	27.991	29.707	32.357	34.764	37.689	49.335	63.167	67.505	71.420	76.154	79.490	86.661
60	35.534	37.485	40.482	43.188	46.459	59.335	74.397	79.082	83.298	88.379	91.952	99.607
70	43.275	45.442	48.758	51.739	55.329	69.334	85.527	90.531	95.023	100.43	104.21	112.32
80	51.172	53.540	57.153	60.391	64.278	79.334	96.578	101.88	106.63	112.33	116.32	124.84
90	59.196	61.754	65.647	69.126	73.291	89.334	107.57	113.15	118.14	124.12	128.30	137.21
100	67.328	70.065	74.222	77.929	82.358	99.334	118.50	124.34	129.56	135.81	140.17	149.45
120	83.852	86.923	91.573	95.705	100.62	119.33	140.23	146.57	152.21	158.95	163.65	173.62
150	109.14	112.67	117.98	122.69	128.28	149.33	172.58	179.58	185.80	193.21	198.36	209.26
200	152.24	156.43	162.73	168.28	174.84	199.33	226.02	233.99	241.06	249.45	255.26	267.54
500	422.30	429.39	439.94	449.15	459.93	499.33	540.93	553.13	563.85	576.49	585.21	603.45

4 F distribution

The table below contains the quantiles of the F distribution with ν_1 and ν_2 degrees of freedom. For $0 < p < 1$, the quantile is the value of x for which

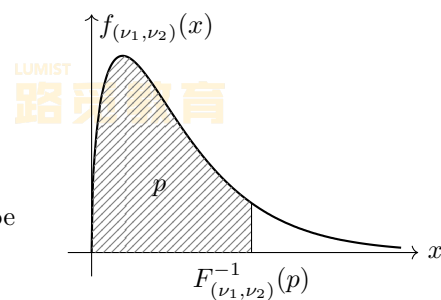
$$\Pr\{X \leq x\} = p, \quad \text{where } X \sim F(\nu_1, \nu_2).$$

Thus

$$x = F_{(\nu_1, \nu_2)}^{-1}(p).$$

The table only contains the quantiles for $p \geq \frac{1}{2}$. For $p < \frac{1}{2}$, quantiles can be obtained by exploiting the symmetry of the F distribution:

$$F_{(\nu_1, \nu_2)}^{-1}(p) = \frac{1}{F_{(\nu_2, \nu_1)}^{-1}(1-p)}.$$



ν_1	ν_2	p				
		0.9	0.95	0.975	0.99	0.995
1	1	39.863	161.45	647.79	4052.2	16211
	2	8.5263	18.513	38.506	98.503	198.50
	3	5.5383	10.128	17.443	34.116	55.552
	4	4.5448	7.7086	12.218	21.198	31.333
	5	4.0604	6.6079	10.007	16.258	22.785
	6	3.7759	5.9874	8.8131	13.745	18.635
	7	3.5894	5.5914	8.0727	12.246	16.236
	8	3.4579	5.3177	7.5709	11.259	14.688
	9	3.3603	5.1174	7.2093	10.561	13.614
	10	3.2850	4.9646	6.9367	10.044	12.826
	11	3.2252	4.8443	6.7241	9.6460	12.226
	12	3.1765	4.7472	6.5538	9.3302	11.754
	13	3.1362	4.6672	6.4143	9.0738	11.374
	14	3.1022	4.6001	6.2979	8.8616	11.060
	15	3.0732	4.5431	6.1995	8.6831	10.798
	16	3.0481	4.4940	6.1151	8.5310	10.575
	17	3.0262	4.4513	6.0420	8.3997	10.384
	18	3.0070	4.4139	5.9781	8.2854	10.218
	19	2.9899	4.3807	5.9216	8.1849	10.073
	20	2.9747	4.3512	5.8715	8.0960	9.9439
1	2	2.9610	4.3248	5.8266	8.0166	9.8295
	3	2.9486	4.3009	5.7863	7.9454	9.7271
	4	2.9374	4.2793	5.7498	7.8811	9.6348
	5	2.9271	4.2597	5.7166	7.8229	9.5513
	6	2.9177	4.2417	5.6864	7.7698	9.4753
	7	2.8807	4.1709	5.5675	7.5625	9.1797
	8	2.8547	4.1213	5.4848	7.4191	8.9763
	9	2.8354	4.0847	5.4239	7.3141	8.8279
	10	2.8205	4.0566	5.3773	7.2339	8.7148
	15	2.8087	4.0343	5.3403	7.1706	8.6258
	20	2.7911	4.0012	5.2856	7.0771	8.4946
	30	2.7786	3.9778	5.2470	7.0114	8.4027
	40	2.7693	3.9604	5.2184	6.9627	8.3346
	50	2.7564	3.9361	5.1786	6.8953	8.2406
	60	2.7478	3.9201	5.1523	6.8509	8.1788
	70	2.7393	3.9042	5.1263	6.8069	8.1177
	80	2.7308	3.8884	5.1004	6.7633	8.0572

→

ν_1	ν_2	p				
		0.9	0.95	0.975	0.99	0.995
1	500	2.7156	3.8601	5.0543	6.6858	7.9498
	+∞	2.7055	3.8415	5.0239	6.6349	7.8794
2	1	49.500	199.50	799.50	4999.5	19999
	2	9.0000	19.000	39.000	99.000	199.00
	3	5.4624	9.5521	16.044	30.817	49.799
	4	4.3246	6.9443	10.649	18.000	26.284
	5	3.7797	5.7861	8.4336	13.274	18.314
	6	3.4633	5.1433	7.2599	10.925	14.544
	7	3.2574	4.7374	6.5415	9.5466	12.404
	8	3.1131	4.4590	6.0595	8.6491	11.042
	9	3.0065	4.2565	5.7147	8.0215	10.107
	10	2.9245	4.1028	5.4564	7.5594	9.4270
	11	2.8595	3.9823	5.2559	7.2057	8.9122
	12	2.8068	3.8853	5.0959	6.9266	8.5096
	13	2.7632	3.8056	4.9653	6.7010	8.1865
	14	2.7265	3.7389	4.8567	6.5149	7.9216
	15	2.6952	3.6823	4.7650	6.3589	7.7008
	16	2.6682	3.6337	4.6867	6.2262	7.5138
	17	2.6446	3.5915	4.6189	6.1121	7.3536
	18	2.6239	3.5546	4.5597	6.0129	7.2148
	19	2.6056	3.5219	4.5075	5.9259	7.0935
	20	2.5893	3.4928	4.4613	5.8489	6.9865
2	21	2.5746	3.4668	4.4199	5.7804	6.8914
	22	2.5613	3.4434	4.3828	5.7190	6.8064
	23	2.5493	3.4221	4.3492	5.6637	6.7300
	24	2.5383	3.4028	4.3187	5.6136	6.6609
	25	2.5283	3.3852	4.2909	5.5680	6.5982
	30	2.4887	3.3158	4.1821	5.3903	6.3547
	35	2.4609	3.2674	4.1065	5.2679	6.1878
	40	2.4404	3.2317	4.0510	5.1785	6.0664
	45	2.4245	3.2043	4.0085	5.1103	5.9741
	50	2.4120	3.1826	3.9749	5.0566	5.9016
	60	2.3933	3.1504	3.9253	4.9774	5.7950
	70	2.3800	3.1277	3.8903	4.9219	5.7204
	80	2.3701	3.1108	3.8643	4.8807	5.6652
	100	2.3564	3.0873	3.8284	4.8239	5.5892
	120	2.3473	3.0718	3.8046	4.7865	5.5393

→

ν_1	ν_2	p				
		0.9	0.95	0.975	0.99	0.995
2	150	2.3383	3.0564	3.7811	4.7495	5.4900
	200	2.3293	3.0411	3.7578	4.7129	5.4412
	500	2.3132	3.0138	3.7162	4.6478	5.3549
	$+\infty$	2.3026	2.9957	3.6889	4.6052	5.2983
3	1	53.593	215.71	864.16	5403.4	21615
	2	9.1618	19.164	39.165	99.166	199.17
	3	5.3908	9.2766	15.439	29.457	47.467
	4	4.1909	6.5914	9.9792	16.694	24.259
	5	3.6195	5.4095	7.7636	12.060	16.530
	6	3.2888	4.7571	6.5988	9.7795	12.917
	7	3.0741	4.3468	5.8898	8.4513	10.882
	8	2.9238	4.0662	5.4160	7.5910	9.5965
	9	2.8129	3.8625	5.0781	6.9919	8.7171
	10	2.7277	3.7083	4.8256	6.5523	8.0807
	11	2.6602	3.5874	4.6300	6.2167	7.6004
	12	2.6055	3.4903	4.4742	5.9525	7.2258
	13	2.5603	3.4105	4.3472	5.7394	6.9258
	14	2.5222	3.3439	4.2417	5.5639	6.6804
	15	2.4898	3.2874	4.1528	5.4170	6.4760
	16	2.4618	3.2389	4.0768	5.2922	6.3034
	17	2.4374	3.1968	4.0112	5.1850	6.1556
	18	2.4160	3.1599	3.9539	5.0919	6.0278
	19	2.3970	3.1274	3.9034	5.0103	5.9161
	20	2.3801	3.0984	3.8587	4.9382	5.8177
	21	2.3649	3.0725	3.8188	4.8740	5.7304
	22	2.3512	3.0491	3.7829	4.8166	5.6524
	23	2.3387	3.0280	3.7505	4.7649	5.5823
	24	2.3274	3.0088	3.7211	4.7181	5.5190
	25	2.3170	2.9912	3.6943	4.6755	5.4615
	30	2.2761	2.9223	3.5894	4.5097	5.2388
	35	2.2474	2.8742	3.5166	4.3957	5.0865
	40	2.2261	2.8387	3.4633	4.3126	4.9758
	45	2.2097	2.8115	3.4224	4.2492	4.8918
	50	2.1967	2.7900	3.3902	4.1993	4.8259
	60	2.1774	2.7581	3.3425	4.1259	4.7290
	70	2.1637	2.7355	3.3090	4.0744	4.6613
	80	2.1535	2.7188	3.2841	4.0363	4.6113
	100	2.1394	2.6955	3.2496	3.9837	4.5424
	120	2.1300	2.6802	3.2269	3.9491	4.4972
	150	2.1207	2.6649	3.2044	3.9149	4.4525
	200	2.1114	2.6498	3.1820	3.8810	4.4084
	500	2.0948	2.6227	3.1423	3.8210	4.3304
	$+\infty$	2.0838	2.6049	3.1161	3.7816	4.2794
4	1	55.833	224.58	899.58	5624.6	22500
	2	9.2434	19.247	39.248	99.249	199.25
	3	5.3426	9.1172	15.101	28.710	46.195
	4	4.1072	6.3882	9.6045	15.977	23.155
	5	3.5202	5.1922	7.3879	11.392	15.556
	6	3.1808	4.5337	6.2272	9.1483	12.028
	7	2.9605	4.1203	5.5226	7.8466	10.050
	8	2.8064	3.8379	5.0526	7.0061	8.8051

→

ν_1	ν_2	p				
		0.9	0.95	0.975	0.99	0.995
4	9	2.6927	3.6331	4.7181	6.4221	7.9559
	10	2.6053	3.4780	4.4683	5.9943	7.3428
	11	2.5362	3.3567	4.2751	5.6683	6.8809
	12	2.4801	3.2592	4.1212	5.4120	6.5211
	13	2.4337	3.1791	3.9959	5.2053	6.2335
	14	2.3947	3.1122	3.8919	5.0354	5.9984
	15	2.3614	3.0556	3.8043	4.8932	5.8029
	16	2.3327	3.0069	3.7294	4.7726	5.6378
	17	2.3077	2.9647	3.6648	4.6690	5.4967
	18	2.2858	2.9277	3.6083	4.5790	5.3746
	19	2.2663	2.8951	3.5587	4.5003	5.2681
	20	2.2489	2.8661	3.5147	4.4307	5.1743
	21	2.2333	2.8401	3.4754	4.3688	5.0911
	22	2.2193	2.8167	3.4401	4.3134	5.0168
	23	2.2065	2.7955	3.4083	4.2636	4.9500
	24	2.1949	2.7763	3.3794	4.2184	4.8898
	25	2.1842	2.7587	3.3530	4.1774	4.8351
	30	2.1422	2.6896	3.2499	4.0179	4.6234
	35	2.1128	2.6415	3.1785	3.9082	4.4788
	40	2.0909	2.6060	3.1261	3.8283	4.3738
	45	2.0742	2.5787	3.0860	3.7674	4.2941
5	50	2.0608	2.5572	3.0544	3.7195	4.2316
	60	2.0410	2.5252	3.0077	3.6490	4.1399
	70	2.0269	2.5027	2.9748	3.5996	4.0758
	80	2.0165	2.4859	2.9504	3.5631	4.0285
	100	2.0019	2.4626	2.9166	3.5127	3.9634
	120	1.9923	2.4472	2.8943	3.4795	3.9207
	150	1.9827	2.4320	2.8722	3.4467	3.8785
	200	1.9732	2.4168	2.8503	3.4143	3.8368
	500	1.9561	2.3898	2.8114	3.3569	3.7632
	$+\infty$	1.9449	2.3719	2.7858	3.3192	3.7151
	1	57.240	230.16	921.85	5763.6	23056
	2	9.2926	19.296	39.298	99.299	199.30
	3	5.3092	9.0135	14.885	28.237	45.392
	4	4.0506	6.2561	9.3645	15.522	22.456
	5	3.4530	5.0503	7.1464	10.967	14.940
	6	3.1075	4.3874	5.9876	8.7459	11.464
	7	2.8833	3.9715	5.2852	7.4604	9.5221
	8	2.7264	3.6875	4.8173	6.6318	8.3018
	9	2.6106	3.4817	4.4844	6.0569	7.4712
	10	2.5216	3.3258	4.2361	5.6363	6.8724
	11	2.4512	3.2039	4.0440	5.3160	6.4217
	12	2.3940	3.1059	3.8911	5.0643	6.0711
	13	2.3467	3.0254	3.7667	4.8616	5.7910
	14	2.3069	2.9582	3.6634	4.6950	5.5623
	15	2.2730	2.9013	3.5764	4.5556	5.3721
	16	2.2438	2.8524	3.5021	4.4374	5.2117
	17	2.2183	2.8100	3.4379	4.3359	5.0746
	18	2.1958	2.7729	3.3820	4.2479	4.9560
	19	2.1760	2.7401	3.3327	4.1708	4.8526
	20	2.1582	2.7109	3.2891	4.1027	4.7616

→

ν_1	ν_2	p				
		0.9	0.95	0.975	0.99	0.995
5	21	2.1423	2.6848	3.2501	4.0421	4.6809
	22	2.1279	2.6613	3.2151	3.9880	4.6088
	23	2.1149	2.6400	3.1835	3.9392	4.5441
	24	2.1030	2.6207	3.1548	3.8951	4.4857
	25	2.0922	2.6030	3.1287	3.8550	4.4327
	30	2.0492	2.5336	3.0265	3.6990	4.2276
	35	2.0191	2.4851	2.9557	3.5919	4.0876
	40	1.9968	2.4495	2.9037	3.5138	3.9860
	45	1.9796	2.4221	2.8640	3.4544	3.9090
	50	1.9660	2.4004	2.8327	3.4077	3.8486
	60	1.9457	2.3683	2.7863	3.3389	3.7599
	70	1.9313	2.3456	2.7537	3.2907	3.6980
	80	1.9206	2.3287	2.7295	3.2550	3.6524
	100	1.9057	2.3053	2.6961	3.2059	3.5895
	120	1.8959	2.2899	2.6740	3.1735	3.5482
	150	1.8861	2.2745	2.6521	3.1416	3.5075
	200	1.8763	2.2592	2.6304	3.1100	3.4674
	500	1.8588	2.2320	2.5919	3.0540	3.3963
	$+\infty$	1.8473	2.2141	2.5665	3.0173	3.3499
6	1	58.204	233.99	937.11	5859.0	23437
	2	9.3255	19.330	39.331	99.333	199.33
	3	5.2847	8.9406	14.735	27.911	44.838
	4	4.0097	6.1631	9.1973	15.207	21.975
	5	3.4045	4.9503	6.9777	10.672	14.513
	6	3.0546	4.2839	5.8198	8.4661	11.073
	7	2.8274	3.8660	5.1186	7.1914	9.1553
	8	2.6683	3.5806	4.6517	6.3707	7.9520
	9	2.5509	3.3738	4.3197	5.8018	7.1339
	10	2.4606	3.2172	4.0721	5.3858	6.5446
	11	2.3891	3.0946	3.8807	5.0692	6.1016
	12	2.3310	2.9961	3.7283	4.8206	5.7570
	13	2.2830	2.9153	3.6043	4.6204	5.4819
	14	2.2426	2.8477	3.5014	4.4558	5.2574
	15	2.2081	2.7905	3.4147	4.3183	5.0708
	16	2.1783	2.7413	3.3406	4.2016	4.9134
	17	2.1524	2.6987	3.2767	4.1015	4.7789
	18	2.1296	2.6613	3.2209	4.0146	4.6627
	19	2.1094	2.6283	3.1718	3.9386	4.5614
	20	2.0913	2.5990	3.1283	3.8714	4.4721
	21	2.0751	2.5727	3.0895	3.8117	4.3931
	22	2.0605	2.5491	3.0546	3.7583	4.3225
	23	2.0472	2.5277	3.0232	3.7102	4.2591
	24	2.0351	2.5082	2.9946	3.6667	4.2019
	25	2.0241	2.4904	2.9685	3.6272	4.1500
	30	1.9803	2.4205	2.8667	3.4735	3.9492
	35	1.9496	2.3718	2.7961	3.3679	3.8123
	40	1.9269	2.3359	2.7444	3.2910	3.7129
	45	1.9094	2.3083	2.7048	3.2325	3.6376
	50	1.8954	2.2864	2.6736	3.1864	3.5785
	60	1.8747	2.2541	2.6274	3.1187	3.4918
	70	1.8600	2.2312	2.5949	3.0712	3.4313
	80	1.8491	2.2142	2.5708	3.0361	3.3867

→

ν_1	ν_2	p				
		0.9	0.95	0.975	0.99	0.995
6	100	1.8339	2.1906	2.5374	2.9877	3.3252
	120	1.8238	2.1750	2.5154	2.9559	3.2849
	150	1.8138	2.1595	2.4936	2.9244	3.2452
	200	1.8038	2.1441	2.4720	2.8933	3.2059
	500	1.7859	2.1167	2.4335	2.8381	3.1366
	$+\infty$	1.7741	2.0986	2.4082	2.8020	3.0913
7	1	58.906	236.77	948.22	5928.4	23715
	2	9.3491	19.353	39.355	99.356	199.36
	3	5.2662	8.8867	14.624	27.672	44.434
	4	3.9790	6.0942	9.0741	14.976	21.622
	5	3.3679	4.8759	6.8531	10.456	14.200
	6	3.0145	4.2067	5.6955	8.2600	10.786
	7	2.7849	3.7870	4.9949	6.9928	8.8854
	8	2.6241	3.5005	4.5286	6.1776	7.6941
	9	2.5053	3.2927	4.1970	5.6129	6.8849
	10	2.4140	3.1355	3.9498	5.2001	6.3025
	11	2.3416	3.0123	3.7586	4.8861	5.8648
	12	2.2828	2.9134	3.6065	4.6395	5.5245
	13	2.2341	2.8321	3.4827	4.4410	5.2529
	14	2.1931	2.7642	3.3799	4.2779	5.0313
	15	2.1582	2.7066	3.2934	4.1415	4.8473
	16	2.1280	2.6572	3.2194	4.0259	4.6920
	17	2.1017	2.6143	3.1556	3.9267	4.5594
	18	2.0785	2.5767	3.0999	3.8406	4.4448
	19	2.0580	2.5435	3.0509	3.7653	4.3448
	20	2.0397	2.5140	3.0074	3.6987	4.2569
	21	2.0233	2.4876	2.9686	3.6396	4.1789
	22	2.0084	2.4638	2.9338	3.5867	4.1094
	23	1.9949	2.4422	2.9023	3.5390	4.0469
	24	1.9826	2.4226	2.8738	3.4959	3.9905
	25	1.9714	2.4047	2.8478	3.4568	3.9394
	30	1.9269	2.3343	2.7460	3.3045	3.7416
	35	1.8957	2.2852	2.6755	3.2000	3.6066
	40	1.8725	2.2490	2.6238	3.1238	3.5088
	45	1.8547	2.2212	2.5842	3.0658	3.4346
	50	1.8405	2.1992	2.5530	3.0202	3.3765
	60	1.8194	2.1665	2.5068	2.9530	3.2911
	70	1.8044	2.1435	2.4743	2.9060	3.2315
	80	1.7933	2.1263	2.4502	2.8713	3.1876
	100	1.7778	2.1025	2.4168	2.8233	3.1271
	120	1.7675	2.0868	2.3948	2.7918	3.0874
	150	1.7572	2.0711	2.3730	2.7606	3.0483
	200	1.7470	2.0556	2.3513	2.7298	3.0097
	500	1.7288	2.0279	2.3129	2.6751	2.9414
	$+\infty$	1.7167	2.0096	2.2875	2.6393	2.8968
8	1	59.439	238.88	956.66	5981.1	23925
	2	9.3668	19.371	39.373	99.374	199.37
	3	5.2517	8.8452	14.540	27.489	44.126
	4	3.9549	6.0410	8.9796	14.799	21.352
	5	3.3393	4.8183	6.7572	10.289	13.961
	6	2.9830	4.1468	5.5996	8.1017	10.566
	7	2.7516	3.7257	4.8993	6.8400	8.6781

→

ν_1	ν_2	p				
		0.9	0.95	0.975	0.99	0.995
8	8	2.5893	3.4381	4.4333	6.0289	7.4959
	9	2.4694	3.2296	4.1020	5.4671	6.6933
	10	2.3772	3.0717	3.8549	5.0567	6.1159
	11	2.3040	2.9480	3.6638	4.7445	5.6821
	12	2.2446	2.8486	3.5118	4.4994	5.3451
	13	2.1953	2.7669	3.3880	4.3021	5.0761
	14	2.1539	2.6987	3.2853	4.1399	4.8566
	15	2.1185	2.6408	3.1987	4.0045	4.6744
	16	2.0880	2.5911	3.1248	3.8896	4.5207
	17	2.0613	2.5480	3.0610	3.7910	4.3894
	18	2.0379	2.5102	3.0053	3.7054	4.2759
	19	2.0171	2.4768	2.9563	3.6305	4.1770
	20	1.9985	2.4471	2.9128	3.5644	4.0900
	21	1.9819	2.4205	2.8740	3.5056	4.0128
	22	1.9668	2.3965	2.8392	3.4530	3.9440
	23	1.9531	2.3748	2.8077	3.4057	3.8822
	24	1.9407	2.3551	2.7791	3.3629	3.8264
	25	1.9292	2.3371	2.7531	3.3239	3.7758
	30	1.8841	2.2662	2.6513	3.1726	3.5801
	35	1.8524	2.2167	2.5807	3.0687	3.4466
	40	1.8289	2.1802	2.5289	2.9930	3.3498
	45	1.8107	2.1521	2.4892	2.9353	3.2764
	50	1.7963	2.1299	2.4579	2.8900	3.2189
	60	1.7748	2.0970	2.4117	2.8233	3.1344
	70	1.7596	2.0737	2.3791	2.7765	3.0755
	80	1.7483	2.0564	2.3549	2.7420	3.0320
	100	1.7324	2.0323	2.3215	2.6943	2.9722
	120	1.7220	2.0164	2.2994	2.6629	2.9330
	150	1.7115	2.0006	2.2775	2.6319	2.8942
	200	1.7011	1.9849	2.2558	2.6012	2.8560
	500	1.6825	1.9569	2.2172	2.5469	2.7885
	$+\infty$	1.6702	1.9384	2.1918	2.5113	2.7444

9	1	59.858	240.54	963.28	6022.5	24091
	2	9.3805	19.385	39.387	99.388	199.39
	3	5.2400	8.8123	14.473	27.345	43.882
	4	3.9357	5.9988	8.9047	14.659	21.139
	5	3.3163	4.7725	6.6811	10.158	13.772
	6	2.9577	4.0990	5.5234	7.9761	10.391
	7	2.7247	3.6767	4.8232	6.7188	8.5138
	8	2.5612	3.3881	4.3572	5.9106	7.3386
	9	2.4403	3.1789	4.0260	5.3511	6.5411
	10	2.3473	3.0204	3.7790	4.9424	5.9676
	11	2.2735	2.8962	3.5879	4.6315	5.5368
	12	2.2135	2.7964	3.4358	4.3875	5.2021
	13	2.1638	2.7144	3.3120	4.1911	4.9351
	14	2.1220	2.6458	3.2093	4.0297	4.7173
	15	2.0862	2.5876	3.1227	3.8948	4.5364
	16	2.0553	2.5377	3.0488	3.7804	4.3838
	17	2.0284	2.4943	2.9849	3.6822	4.2535
	18	2.0047	2.4563	2.9291	3.5971	4.1410
	19	1.9836	2.4227	2.8801	3.5225	4.0428

→

ν_1	ν_2	p				
		0.9	0.95	0.975	0.99	0.995
9	20	1.9649	2.3928	2.8365	3.4567	3.9564
	21	1.9480	2.3660	2.7977	3.3981	3.8799
	22	1.9327	2.3419	2.7628	3.3458	3.8116
	23	1.9189	2.3201	2.7313	3.2986	3.7502
	24	1.9063	2.3002	2.7027	3.2560	3.6949
	25	1.8947	2.2821	2.6766	3.2172	3.6447
	30	1.8490	2.2107	2.5746	3.0665	3.4505
	35	1.8168	2.1608	2.5039	2.9630	3.3180
	40	1.7929	2.1240	2.4519	2.8876	3.2220
	45	1.7745	2.0958	2.4122	2.8301	3.1492
	50	1.7598	2.0734	2.3808	2.7850	3.0920
	60	1.7380	2.0401	2.3344	2.7185	3.0083
	70	1.7225	2.0166	2.3017	2.6719	2.9498
	80	1.7110	1.9991	2.2775	2.6374	2.9066
	100	1.6949	1.9748	2.2439	2.5898	2.8472
	120	1.6842	1.9588	2.2217	2.5586	2.8083
	150	1.6736	1.9428	2.1998	2.5277	2.7698
	200	1.6630	1.9269	2.1780	2.4971	2.7319
	500	1.6441	1.8986	2.1392	2.4429	2.6649
	$+\infty$	1.6315	1.8799	2.1136	2.4073	2.6210

10	1	60.195	241.88	968.63	6055.8	24224
	2	9.3916	19.396	39.398	99.399	199.40
	3	5.2304	8.7855	14.419	27.229	43.686
	4	3.9199	5.9644	8.8439	14.546	20.967
	5	3.2974	4.7351	6.6192	10.051	13.618
	6	2.9369	4.0600	5.4613	7.8741	10.250
	7	2.7025	3.6365	4.7611	6.6201	8.3803
	8	2.5380	3.3472	4.2951	5.8143	7.2106
	9	2.4163	3.1373	3.9639	5.2565	6.4172
	10	2.3226	2.9782	3.7168	4.8491	5.8467
	11	2.2482	2.8536	3.5257	4.5393	5.4183
	12	2.1878	2.7534	3.3736	4.2961	5.0855
	13	2.1376	2.6710	3.2497	4.1003	4.8199
	14	2.0954	2.6022	3.1469	3.9394	4.6034
	15	2.0593	2.5437	3.0602	3.8049	4.4235
	16	2.0281	2.4935	2.9862	3.6909	4.2719
	17	2.0009	2.4499	2.9222	3.5931	4.1424
	18	1.9770	2.4117	2.8664	3.5082	4.0305
	19	1.9557	2.3779	2.8172	3.4338	3.9329
	20	1.9367	2.3479	2.7737	3.3682	3.8470
10	21	1.9197	2.3210	2.7348	3.3098	3.7709
	22	1.9043	2.2967	2.6998	3.2576	3.7030
	23	1.8903	2.2747	2.6682	3.2106	3.6420
	24	1.8775	2.2547	2.6396	3.1681	3.5870
	25	1.8658	2.2365	2.6135	3.1294	3.5370
	30	1.8195	2.1646	2.5112	2.9791	3.3440
	35	1.7869	2.1143	2.4403	2.8758	3.2123
	40	1.7627	2.0772	2.3882	2.8005	3.1167
	45	1.7440	2.0487	2.3483	2.7432	3.0443
	50	1.7291	2.0261	2.3168	2.6981	2.9875
	60	1.7070	1.9926	2.2702	2.6318	2.9042

→

ν_1	ν_2	p				
		0.9	0.95	0.975	0.99	0.995
10	70	1.6913	1.9689	2.2374	2.5852	2.8460
	80	1.6796	1.9512	2.2130	2.5508	2.8031
	100	1.6632	1.9267	2.1793	2.5033	2.7440
	120	1.6524	1.9105	2.1570	2.4721	2.7052
	150	1.6416	1.8943	2.1349	2.4412	2.6669
	200	1.6308	1.8783	2.1130	2.4106	2.6292
	500	1.6115	1.8496	2.0740	2.3565	2.5625
	$+\infty$	1.5987	1.8307	2.0483	2.3209	2.5188
11	1	60.473	242.98	973.03	6083.3	24334
	2	9.4006	19.405	39.407	99.408	199.41
	3	5.2224	8.7633	14.374	27.133	43.524
	4	3.9067	5.9358	8.7935	14.452	20.824
	5	3.2816	4.7040	6.5678	9.9626	13.491
	6	2.9195	4.0274	5.4098	7.7896	10.133
	7	2.6839	3.6030	4.7095	6.5382	8.2697
	8	2.5186	3.3130	4.2434	5.7343	7.1045
	9	2.3961	3.1025	3.9121	5.1779	6.3142
	10	2.3018	2.9430	3.6649	4.7715	5.7462
	11	2.2269	2.8179	3.4737	4.4624	5.3197
	12	2.1660	2.7173	3.3215	4.2198	4.9884
	13	2.1155	2.6347	3.1975	4.0245	4.7240
	14	2.0729	2.5655	3.0946	3.8640	4.5085
	15	2.0366	2.5068	3.0078	3.7299	4.3295
	16	2.0051	2.4564	2.9337	3.6162	4.1785
	17	1.9777	2.4126	2.8696	3.5185	4.0496
	18	1.9535	2.3742	2.8137	3.4338	3.9382
	19	1.9321	2.3402	2.7645	3.3596	3.8410
	20	1.9129	2.3100	2.7209	3.2941	3.7555
	21	1.8956	2.2829	2.6819	3.2359	3.6798
	22	1.8801	2.2585	2.6469	3.1837	3.6122
	23	1.8659	2.2364	2.6152	3.1368	3.5515
	24	1.8530	2.2163	2.5865	3.0944	3.4967
	25	1.8412	2.1979	2.5603	3.0558	3.4470
	30	1.7944	2.1256	2.4577	2.9057	3.2547
	35	1.7614	2.0750	2.3866	2.8026	3.1236
	40	1.7369	2.0376	2.3343	2.7274	3.0284
	45	1.7180	2.0088	2.2943	2.6701	2.9563
	50	1.7029	1.9861	2.2627	2.6250	2.8997
	60	1.6805	1.9522	2.2159	2.5587	2.8166
	70	1.6645	1.9283	2.1829	2.5122	2.7587
	80	1.6526	1.9105	2.1584	2.4777	2.7159
	100	1.6360	1.8857	2.1245	2.4302	2.6570
	120	1.6250	1.8693	2.1021	2.3990	2.6183
	150	1.6140	1.8530	2.0799	2.3681	2.5802
	200	1.6031	1.8368	2.0578	2.3375	2.5425
	500	1.5835	1.8078	2.0186	2.2833	2.4760
	$+\infty$	1.5705	1.7886	1.9927	2.2477	2.4324
12	1	60.705	243.91	976.71	6106.3	24426
	2	9.4081	19.413	39.415	99.416	199.42
	3	5.2156	8.7446	14.337	27.052	43.387
	4	3.8955	5.9117	8.7512	14.374	20.705
	5	3.2682	4.6777	6.5245	9.8883	13.384

→

ν_1	ν_2	p				
		0.9	0.95	0.975	0.99	0.995
12	6	2.9047	3.9999	5.3662	7.7183	10.034
	7	2.6681	3.5747	4.6658	6.4691	8.1764
	8	2.5020	3.2839	4.1997	5.6667	7.0149
	9	2.3789	3.0729	3.8682	5.1114	6.2274
	10	2.2841	2.9130	3.6209	4.7059	5.6613
	11	2.2087	2.7876	3.4296	4.3974	5.2363
	12	2.1474	2.6866	3.2773	4.1553	4.9062
	13	2.0966	2.6037	3.1532	3.9603	4.6429
	14	2.0537	2.5342	3.0502	3.8001	4.4281
	15	2.0171	2.4753	2.9633	3.6662	4.2497
	16	1.9854	2.4247	2.8890	3.5527	4.0994
	17	1.9577	2.3807	2.8249	3.4552	3.9709
	18	1.9333	2.3421	2.7689	3.3706	3.8599
	19	1.9117	2.3080	2.7196	3.2965	3.7631
	20	1.8924	2.2776	2.6758	3.2311	3.6779
	21	1.8750	2.2504	2.6368	3.1730	3.6024
	22	1.8593	2.2258	2.6017	3.1209	3.5350
	23	1.8450	2.2036	2.5699	3.0740	3.4745
	24	1.8319	2.1834	2.5411	3.0316	3.4199
	25	1.8200	2.1649	2.5149	2.9931	3.3704
	30	1.7727	2.0921	2.4120	2.8431	3.1787
	35	1.7394	2.0411	2.3406	2.7400	3.0480
	40	1.7146	2.0035	2.2882	2.6648	2.9531
	45	1.6954	1.9745	2.2480	2.6076	2.8811
	50	1.6802	1.9515	2.2162	2.5625	2.8247
	60	1.6574	1.9174	2.1692	2.4961	2.7419
	70	1.6413	1.8932	2.1361	2.4496	2.6840
	80	1.6292	1.8753	2.1115	2.4151	2.6413
	100	1.6124	1.8503	2.0773	2.3676	2.5825
	120	1.6012	1.8337	2.0548	2.3363	2.5439
	150	1.5901	1.8172	2.0325	2.3053	2.5059
	200	1.5789	1.8008	2.0103	2.2747	2.4683
	500	1.5590	1.7715	1.9708	2.2204	2.4018
	$+\infty$	1.5458	1.7522	1.9447	2.1847	2.3583
13	1	60.903	244.69	979.84	6125.9	24505
	2	9.4145	19.419	39.421	99.422	199.42
	3	5.2098	8.7287	14.304	26.983	43.271
	4	3.8859	5.8911	8.7150	14.307	20.603
	5	3.2567	4.6552	6.4876	9.8248	13.293
	6	2.8920	3.9764	5.3290	7.6575	9.9501
	7	2.6545	3.5503	4.6285	6.4100	8.0967
	8	2.4876	3.2590	4.1622	5.6089	6.9384
	9	2.3640	3.0475	3.8306	5.0545	6.1530
	10	2.2687	2.8872	3.5832	4.6496	5.5887
	11	2.1930	2.7614	3.3917	4.3416	5.1649
	12	2.1313	2.6602	3.2393	4.0999	4.8358
	13	2.0802	2.5769	3.1150	3.9052	4.5733
	14	2.0370	2.5073	3.0119	3.7452	4.3591
	15	2.0001	2.4481	2.9249	3.6115	4.1813
	16	1.9682	2.3973	2.8506	3.4981	4.0314
	17	1.9404	2.3531	2.7863	3.4007	3.9033
	18	1.9158	2.3143	2.7302	3.3162	3.7926

→

ν_1	ν_2	p				
		0.9	0.95	0.975	0.99	0.995
13	19	1.8940	2.2800	2.6808	3.2422	3.6961
	20	1.8745	2.2495	2.6369	3.1769	3.6111
	21	1.8570	2.2222	2.5978	3.1187	3.5358
	22	1.8411	2.1975	2.5626	3.0667	3.4686
	23	1.8267	2.1752	2.5308	3.0199	3.4083
	24	1.8136	2.1548	2.5019	2.9775	3.3538
	25	1.8015	2.1362	2.4756	2.9389	3.3044
	30	1.7538	2.0630	2.3724	2.7890	3.1132
	35	1.7201	2.0117	2.3008	2.6859	2.9827
	40	1.6950	1.9738	2.2481	2.6107	2.8880
	45	1.6757	1.9446	2.2078	2.5534	2.8162
	50	1.6602	1.9214	2.1758	2.5083	2.7599
	60	1.6372	1.8870	2.1286	2.4419	2.6771
	70	1.6209	1.8627	2.0953	2.3953	2.6193
	80	1.6086	1.8445	2.0706	2.3608	2.5767
	100	1.5916	1.8193	2.0363	2.3132	2.5180
	120	1.5803	1.8026	2.0136	2.2818	2.4794
	150	1.5690	1.7859	1.9911	2.2508	2.4413
	200	1.5577	1.7694	1.9688	2.2201	2.4038
	500	1.5374	1.7398	1.9290	2.1656	2.3373
	$+\infty$	1.5240	1.7202	1.9027	2.1299	2.2938
14	1	61.073	245.36	982.53	6142.7	24572
	2	9.4200	19.424	39.427	99.428	199.43
	3	5.2047	8.7149	14.277	26.924	43.172
	4	3.8776	5.8733	8.6838	14.249	20.515
	5	3.2468	4.6358	6.4556	9.7700	13.215
	6	2.8809	3.9559	5.2968	7.6049	9.8774
	7	2.6426	3.5292	4.5961	6.3590	8.0279
	8	2.4752	3.2374	4.1297	5.5589	6.8721
	9	2.3510	3.0255	3.7980	5.0052	6.0887
	10	2.2553	2.8647	3.5504	4.6008	5.5257
	11	2.1792	2.7386	3.3588	4.2932	5.1031
	12	2.1173	2.6371	3.2062	4.0518	4.7748
	13	2.0658	2.5536	3.0819	3.8573	4.5129
	14	2.0224	2.4837	2.9786	3.6975	4.2993
	15	1.9853	2.4244	2.8915	3.5639	4.1219
	16	1.9532	2.3733	2.8170	3.4506	3.9723
	17	1.9252	2.3290	2.7526	3.3533	3.8445
	18	1.9004	2.2900	2.6964	3.2689	3.7341
	19	1.8785	2.2556	2.6469	3.1949	3.6378
	20	1.8588	2.2250	2.6030	3.1296	3.5530
	21	1.8412	2.1975	2.5638	3.0715	3.4779
	22	1.8252	2.1727	2.5285	3.0195	3.4108
	23	1.8107	2.1502	2.4966	2.9727	3.3506
	24	1.7974	2.1298	2.4677	2.9303	3.2962
	25	1.7853	2.1111	2.4413	2.8917	3.2469
	30	1.7371	2.0374	2.3378	2.7418	3.0560
	35	1.7031	1.9858	2.2659	2.6387	2.9258
	40	1.6778	1.9476	2.2130	2.5634	2.8312
	45	1.6582	1.9182	2.1725	2.5060	2.7595
	50	1.6426	1.8949	2.1404	2.4609	2.7032

→

ν_1	ν_2	p				
		0.9	0.95	0.975	0.99	0.995
14	60	1.6193	1.8602	2.0929	2.3943	2.6205
	70	1.6028	1.8357	2.0595	2.3477	2.5627
	80	1.5904	1.8174	2.0346	2.3131	2.5201
	100	1.5731	1.7919	2.0001	2.2654	2.4614
	120	1.5617	1.7750	1.9773	2.2339	2.4228
	150	1.5502	1.7582	1.9546	2.2028	2.3847
	200	1.5388	1.7415	1.9322	2.1721	2.3472
	500	1.5182	1.7116	1.8921	2.1174	2.2806
	$+\infty$	1.5046	1.6918	1.8656	2.0815	2.2371
	15	1	61.220	245.95	984.87	6157.3
15	2	9.4247	19.429	39.431	99.433	199.43
	3	5.2003	8.7029	14.253	26.872	43.085
	4	3.8704	5.8578	8.6565	14.198	20.438
	5	3.2380	4.6188	6.4277	9.7222	13.146
	6	2.8712	3.9381	5.2687	7.5590	9.8140
	7	2.6322	3.5107	4.5678	6.3143	7.9678
	8	2.4642	3.2184	4.1012	5.5151	6.8143
	9	2.3396	3.0061	3.7694	4.9621	6.0325
	10	2.2435	2.8450	3.5217	4.5581	5.4707
	11	2.1671	2.7186	3.3299	4.2509	5.0489
	12	2.1049	2.6169	3.1772	4.0096	4.7213
	13	2.0532	2.5331	3.0527	3.8154	4.4600
	14	2.0095	2.4630	2.9493	3.6557	4.2468
	15	1.9722	2.4034	2.8621	3.5222	4.0698
	16	1.9399	2.3522	2.7875	3.4089	3.9205
	17	1.9117	2.3077	2.7230	3.3117	3.7929
	18	1.8868	2.2686	2.6667	3.2273	3.6827
	19	1.8647	2.2341	2.6171	3.1533	3.5866
	20	1.8449	2.2033	2.5731	3.0880	3.5020
	21	1.8271	2.1757	2.5338	3.0300	3.4270
	22	1.8111	2.1508	2.4984	2.9779	3.3600
	23	1.7964	2.1282	2.4665	2.9311	3.2999
	24	1.7831	2.1077	2.4374	2.8887	3.2456
	25	1.7708	2.0889	2.4110	2.8502	3.1963
	30	1.7223	2.0148	2.3072	2.7002	3.0057
	35	1.6880	1.9629	2.2350	2.5970	2.8756
	40	1.6624	1.9245	2.1819	2.5216	2.7811
	45	1.6426	1.8949	2.1412	2.4642	2.7094
	50	1.6269	1.8714	2.1090	2.4190	2.6531
	60	1.6034	1.8364	2.0613	2.3523	2.5705
	70	1.5866	1.8117	2.0277	2.3055	2.5127
	80	1.5741	1.7932	2.0026	2.2709	2.4700
	100	1.5566	1.7675	1.9679	2.2230	2.4113
	120	1.5450	1.7505	1.9450	2.1915	2.3727
	150	1.5334	1.7335	1.9222	2.1603	2.3346
	200	1.5218	1.7166	1.8996	2.1294	2.2970
	500	1.5010	1.6864	1.8592	2.0746	2.2304
	$+\infty$	1.4871	1.6664	1.8326	2.0385	2.1868
16	1	61.350	246.46	986.92	6170.1	24681
	2	9.4289	19.433	39.435	99.437	199.44
	3	5.1964	8.6923	14.232	26.827	43.008

→

ν_1	ν_2	p				
		0.9	0.95	0.975	0.99	0.995
16	4	3.8639	5.8441	8.6326	14.154	20.371
	5	3.2303	4.6038	6.4032	9.6802	13.086
	6	2.8626	3.9223	5.2439	7.5186	9.7582
	7	2.6230	3.4944	4.5428	6.2750	7.9148
	8	2.4545	3.2016	4.0761	5.4766	6.7633
	9	2.3295	2.9890	3.7441	4.9240	5.9829
	10	2.2330	2.8276	3.4963	4.5204	5.4221
	11	2.1563	2.7009	3.3044	4.2134	5.0011
	12	2.0938	2.5989	3.1515	3.9724	4.6741
	13	2.0419	2.5149	3.0269	3.7783	4.4132
	14	1.9981	2.4446	2.9234	3.6187	4.2005
	15	1.9605	2.3849	2.8360	3.4852	4.0237
	16	1.9281	2.3335	2.7614	3.3720	3.8747
	17	1.8997	2.2888	2.6968	3.2748	3.7473
	18	1.8747	2.2496	2.6404	3.1904	3.6373
	19	1.8524	2.2149	2.5907	3.1165	3.5412
	20	1.8325	2.1840	2.5465	3.0512	3.4568
	21	1.8146	2.1563	2.5071	2.9931	3.3818
	22	1.7984	2.1313	2.4717	2.9411	3.3150
	23	1.7837	2.1086	2.4396	2.8943	3.2549
	24	1.7703	2.0880	2.4105	2.8519	3.2007
	25	1.7579	2.0691	2.3840	2.8133	3.1515
	30	1.7090	1.9946	2.2799	2.6632	2.9611
	35	1.6744	1.9424	2.2075	2.5599	2.8310
	40	1.6486	1.9037	2.1542	2.4844	2.7365
	45	1.6287	1.8740	2.1133	2.4269	2.6648
	50	1.6128	1.8503	2.0810	2.3816	2.6086
	60	1.5890	1.8151	2.0330	2.3148	2.5259
	70	1.5721	1.7902	1.9992	2.2679	2.4681
	80	1.5594	1.7716	1.9741	2.2332	2.4254
	100	1.5418	1.7456	1.9391	2.1852	2.3666
	120	1.5300	1.7285	1.9161	2.1536	2.3280
	150	1.5182	1.7113	1.8931	2.1223	2.2898
	200	1.5065	1.6943	1.8704	2.0913	2.2521
	500	1.4854	1.6638	1.8297	2.0362	2.1854
	$+\infty$	1.4714	1.6435	1.8028	2.0000	2.1417

ν_1	ν_2	p				
		0.9	0.95	0.975	0.99	0.995
17	17	1.8889	2.2719	2.6733	3.2419	3.7066
	18	1.8638	2.2325	2.6168	3.1575	3.5967
	19	1.8414	2.1977	2.5670	3.0836	3.5008
	20	1.8214	2.1667	2.5228	3.0183	3.4164
	21	1.8034	2.1389	2.4833	2.9602	3.3416
	22	1.7871	2.1138	2.4478	2.9082	3.2748
	23	1.7723	2.0910	2.4157	2.8613	3.2148
	24	1.7587	2.0703	2.3865	2.8189	3.1606
	25	1.7463	2.0513	2.3599	2.7803	3.1114
	30	1.6970	1.9765	2.2554	2.6301	2.9211
	35	1.6622	1.9240	2.1828	2.5266	2.7911
	40	1.6362	1.8851	2.1293	2.4511	2.6966
	45	1.6161	1.8551	2.0883	2.3935	2.6249
	50	1.6000	1.8313	2.0558	2.3481	2.5686
	60	1.5760	1.7959	2.0076	2.2811	2.4859
	70	1.5589	1.7708	1.9736	2.2341	2.4281
	80	1.5461	1.7520	1.9483	2.1993	2.3854
	100	1.5283	1.7259	1.9132	2.1511	2.3265
	120	1.5164	1.7085	1.8900	2.1194	2.2878
	150	1.5045	1.6913	1.8669	2.0880	2.2496
	200	1.4926	1.6741	1.8440	2.0569	2.2118
	500	1.4712	1.6432	1.8030	2.0016	2.1449
	$+\infty$	1.4570	1.6228	1.7759	1.9652	2.1011

18	1	61.566	247.32	990.35	6191.5	24767
	2	9.4358	19.440	39.442	99.444	199.44
	3	5.1898	8.6745	14.196	26.751	42.880
	4	3.8531	5.8211	8.5924	14.080	20.258
	5	3.2172	4.5785	6.3619	9.6096	12.985
	6	2.8481	3.8957	5.2021	7.4507	9.6644
	7	2.6074	3.4669	4.5008	6.2089	7.8258
	8	2.4380	3.1733	4.0338	5.4116	6.6775
	9	2.3123	2.9600	3.7015	4.8599	5.8994
	10	2.2153	2.7980	3.4534	4.4569	5.3403
	11	2.1380	2.6709	3.2612	4.1503	4.9205
	12	2.0750	2.5684	3.1081	3.9095	4.5945
	13	2.0227	2.4841	2.9832	3.7156	4.3344
	14	1.9785	2.4134	2.8795	3.5561	4.1221
	15	1.9407	2.3533	2.7919	3.4228	3.9459
	16	1.9079	2.3016	2.7170	3.3096	3.7972
	17	1.8792	2.2567	2.6522	3.2124	3.6701
	18	1.8539	2.2172	2.5956	3.1280	3.5603
	19	1.8314	2.1823	2.5457	3.0541	3.4645
	20	1.8113	2.1511	2.5014	2.9887	3.3802
	21	1.7932	2.1232	2.4618	2.9306	3.3054
	22	1.7768	2.0980	2.4262	2.8786	3.2387
	23	1.7619	2.0751	2.3940	2.8317	3.1787
	24	1.7483	2.0543	2.3648	2.7892	3.1246
	25	1.7358	2.0353	2.3381	2.7506	3.0754
	30	1.6862	1.9601	2.2334	2.6003	2.8852
	35	1.6511	1.9073	2.1605	2.4967	2.7551
	40	1.6249	1.8682	2.1068	2.4210	2.6607
	45	1.6046	1.8381	2.0656	2.3633	2.5889

→

→

ν_1	ν_2	p				
		0.9	0.95	0.975	0.99	0.995
18	50	1.5884	1.8141	2.0330	2.3178	2.5326
	60	1.5642	1.7784	1.9846	2.2507	2.4498
	70	1.5470	1.7531	1.9504	2.2036	2.3919
	80	1.5340	1.7342	1.9250	2.1686	2.3492
	100	1.5160	1.7079	1.8897	2.1203	2.2902
	120	1.5039	1.6904	1.8663	2.0885	2.2514
	150	1.4919	1.6730	1.8431	2.0570	2.2131
	200	1.4799	1.6556	1.8200	2.0257	2.1753
	500	1.4583	1.6245	1.7787	1.9702	2.1082
	$+\infty$	1.4439	1.6038	1.7515	1.9336	2.0642
19	1	61.658	247.69	991.80	6200.6	24803
	2	9.4387	19.443	39.445	99.447	199.45
	3	5.1870	8.6670	14.181	26.719	42.826
	4	3.8485	5.8114	8.5753	14.048	20.210
	5	3.2117	4.5678	6.3444	9.5797	12.942
	6	2.8419	3.8844	5.1844	7.4219	9.6247
	7	2.6008	3.4551	4.4829	6.1808	7.7881
	8	2.4310	3.1613	4.0158	5.3840	6.6411
	9	2.3050	2.9477	3.6833	4.8327	5.8639
	10	2.2077	2.7854	3.4351	4.4299	5.3055
	11	2.1302	2.6581	3.2428	4.1234	4.8863
	12	2.0670	2.5554	3.0896	3.8827	4.5606
	13	2.0145	2.4709	2.9646	3.6888	4.3008
	14	1.9701	2.4000	2.8607	3.5294	4.0888
	15	1.9321	2.3398	2.7730	3.3961	3.9127
	16	1.8992	2.2880	2.6980	3.2829	3.7641
	17	1.8704	2.2429	2.6331	3.1857	3.6372
	18	1.8450	2.2033	2.5764	3.1013	3.5275
	19	1.8224	2.1683	2.5265	3.0274	3.4318
	20	1.8022	2.1370	2.4821	2.9620	3.3475
	21	1.7840	2.1090	2.4424	2.9039	3.2728
	22	1.7675	2.0837	2.4067	2.8518	3.2060
	23	1.7525	2.0608	2.3745	2.8049	3.1461
	24	1.7388	2.0399	2.3452	2.7624	3.0920
	25	1.7263	2.0207	2.3184	2.7238	3.0429
	30	1.6763	1.9452	2.2134	2.5732	2.8526
	35	1.6410	1.8922	2.1403	2.4695	2.7226
	40	1.6146	1.8529	2.0864	2.3937	2.6281
	45	1.5941	1.8226	2.0450	2.3359	2.5563
	50	1.5778	1.7985	2.0122	2.2903	2.4999
	60	1.5534	1.7625	1.9636	2.2230	2.4171
	70	1.5360	1.7371	1.9293	2.1758	2.3591
	80	1.5230	1.7180	1.9037	2.1408	2.3163
	100	1.5047	1.6915	1.8682	2.0923	2.2572
	120	1.4926	1.6739	1.8447	2.0604	2.2183
	150	1.4804	1.6563	1.8213	2.0287	2.1800
	200	1.4683	1.6388	1.7981	1.9973	2.1420
	500	1.4464	1.6074	1.7566	1.9415	2.0748
	$+\infty$	1.4318	1.5865	1.7291	1.9048	2.0306

ν_1	ν_2	p				
		0.9	0.95	0.975	0.99	0.995
20	3	5.1845	8.6602	14.167	26.690	42.778
	4	3.8443	5.8025	8.5599	14.020	20.167
	5	3.2067	4.5581	6.3286	9.5526	12.903
	6	2.8363	3.8742	5.1684	7.3958	9.5888
	7	2.5947	3.4445	4.4667	6.1554	7.7540
	8	2.4246	3.1503	3.9995	5.3591	6.6082
	9	2.2983	2.9365	3.6669	4.8080	5.8318
	10	2.2007	2.7740	3.4185	4.4054	5.2740
	11	2.1230	2.6464	3.2261	4.0990	4.8552
	12	2.0597	2.5436	3.0728	3.8584	4.5299
	13	2.0070	2.4589	2.9477	3.6646	4.2703
	14	1.9625	2.3879	2.8437	3.5052	4.0585
	15	1.9243	2.3275	2.7559	3.3719	3.8826
	16	1.8913	2.2756	2.6808	3.2587	3.7342
	17	1.8624	2.2304	2.6158	3.1615	3.6073
	18	1.8368	2.1906	2.5590	3.0771	3.4977
	19	1.8142	2.1555	2.5089	3.0031	3.4020
	20	1.7938	2.1242	2.4645	2.9377	3.3178
	21	1.7756	2.0960	2.4247	2.8796	3.2431
	22	1.7590	2.0707	2.3890	2.8274	3.1764
	23	1.7439	2.0476	2.3567	2.7805	3.1165
	24	1.7302	2.0267	2.3273	2.7380	3.0624
	25	1.7175	2.0075	2.3005	2.6993	3.0133
	30	1.6673	1.9317	2.1952	2.5487	2.8230
	35	1.6317	1.8784	2.1218	2.4448	2.6930
	40	1.6052	1.8389	2.0677	2.3689	2.5984
	45	1.5846	1.8084	2.0262	2.3109	2.5266
	50	1.5681	1.7841	1.9933	2.2652	2.4702
	60	1.5435	1.7480	1.9445	2.1978	2.3872
	70	1.5259	1.7223	1.9100	2.1504	2.3291
	80	1.5128	1.7032	1.8843	2.1153	2.2862
	100	1.4943	1.6764	1.8486	2.0666	2.2270
	120	1.4821	1.6587	1.8249	2.0346	2.1881
	150	1.4698	1.6410	1.8014	2.0028	2.1496
	200	1.4575	1.6233	1.7780	1.9713	2.1116
	500	1.4354	1.5916	1.7362	1.9152	2.0441
	$+\infty$	1.4206	1.5705	1.7085	1.8783	1.9998

20	1	61.740	248.01	993.10	6208.7	24836
	2	9.4413	19.446	39.448	99.449	199.45

→

5 Non-parametric tests

5.1 Wilcoxon signed-rank test (one sample)

The table below contains **the end points of the rejection regions** for the Wilcoxon signed-rank test for inference about the median η of a symmetric distribution.

Denote by w_- the rank sum of the observations less than η_0 , and denote by w_+ the rank sum of the observations greater than η_0 .

$H_0 : \eta \leq \eta_0$ vs. $H_1 : \eta > \eta_0$ (one-sided): Reject H_0 if w_- is not larger than the tabulated value.
 $H_0 : \eta \geq \eta_0$ vs. $H_1 : \eta < \eta_0$ (one-sided): Reject H_0 if w_+ is not larger than the tabulated value.
 $H_0 : \eta = \eta_0$ vs. $H_1 : \eta \neq \eta_0$ (two-sided): Reject H_0 if $\min\{w_+, w_-\}$ is not larger than the tabulated value.

“–” indicates that H_0 is never rejected for the chosen size α and sample size n .

n	size α (one-sided)				n	size α (one-sided)			
	5%	2.5%	1%	0.5%		5%	2.5%	1%	0.5%
	size α (two-sided)					size α (two-sided)			
	10%	5%	2%	1%		10%	5%	2%	1%
1	—	—	—	—	31	163	147	130	118
2	—	—	—	—	32	175	159	140	128
3	—	—	—	—	33	187	170	151	138
4	—	—	—	—	34	200	182	162	148
5	0	—	—	—	35	213	195	173	159
6	2	0	—	—	36	227	208	185	171
7	3	2	0	—	37	241	221	198	182
8	5	3	1	0	38	256	235	211	194
9	8	5	3	1	39	271	249	224	207
10	10	8	5	3	40	286	264	238	220
11	13	10	7	5	41	302	279	252	233
12	17	13	9	7	42	319	294	266	247
13	21	17	12	9	43	336	310	281	261
14	25	21	15	12	44	353	327	296	276
15	30	25	19	15	45	371	343	312	291
16	35	29	23	19	46	389	361	328	307
17	41	34	27	23	47	407	378	345	322
18	47	40	32	27	48	426	396	362	339
19	53	46	37	32	49	446	415	379	355
20	60	52	43	37	50	466	434	397	373
21	67	58	49	42	51	486	453	416	390
22	75	65	55	48	52	507	473	434	408
23	83	73	62	54	53	529	494	454	427
24	91	81	69	61	54	550	514	473	445
25	100	89	76	68	55	573	536	493	465
26	110	98	84	75	56	595	557	514	484
27	119	107	92	83	57	618	579	535	504
28	130	116	101	91	58	642	602	556	525
29	140	126	110	100	59	666	625	578	546
30	151	137	120	109	60	690	648	600	567

n	size α (one-sided)			
	5%	2.5%	1%	0.5%
	size α (two-sided)			
	10%	5%	2%	1%
31	163	147	130	118
32	175	159	140	128
33	187	170	151	138
34	200	182	162	148
35	213	195	173	159
36	227	208	185	171
37	241	221	198	182
38	256	235	211	194
39	271	249	224	207
40	286	264	238	220
41	302	279	252	233
42	319	294	266	247
43	336	310	281	261
44	353	327	296	276
45	371	343	312	291
46	389	361	328	307
47	407	378	345	322
48	426	396	362	339
49	446	415	379	355
50	466	434	397	373
51	486	453	416	390
52	507	473	434	408
53	529	494	454	427
54	550	514	473	445
55	573	536	493	465
56	595	557	514	484
57	618	579	535	504
58	642	602	556	525
59	666	625	578	546
60	690	648	600	567

n	size α (one-sided)			
	5%	2.5%	1%	0.5%
	size α (two-sided)			
	10%	5%	2%	1%
61	715	672	623	589
62	741	697	646	611
63	767	721	669	634
64	793	747	693	657
65	820	772	718	681
66	847	798	742	705
67	875	825	768	729
68	903	852	793	754
69	931	879	819	779
70	960	907	846	805
71	990	936	873	831
72	1020	964	901	858
73	1050	994	928	884
74	1081	1023	957	912
75	1112	1053	986	940
76	1144	1084	1015	968
77	1176	1115	1044	997
78	1209	1147	1075	1026
79	1242	1179	1105	1056
80	1276	1211	1136	1086
81	1310	1244	1168	1116
82	1345	1277	1200	1147
83	1380	1311	1232	1178
84	1415	1345	1265	1210
85	1451	1380	1298	1242
86	1487	1415	1332	1275
87	1524	1451	1366	1308
88	1561	1487	1400	1342
89	1599	1523	1435	1376
90	1638	1560	1471	1410

5.2 Mann–Whitney U/Wilcoxon rank sum test (two sample)

The table below contains **the end points of the rejection regions** for the Mann–Whitney test for inference about the medians η_1 and η_2 of two translated symmetric distributions based on two samples of size n_1 and n_2 , respectively.

Denote by r_1 and r_2 the rank sums obtained for the observations from the first and the second sample.

Define

$$u_1 = r_1 - \frac{n_1(n_1 + 1)}{2}, \quad u_2 = r_2 - \frac{n_2(n_2 + 1)}{2}.$$

$H_0 : \eta_1 \leq \eta_2$ vs. $H_1 : \eta_1 > \eta_2$ (one-sided): Reject H_0 if u_2 is not larger than the tabulated value.

$H_0 : \eta_1 \geq \eta_2$ vs. $H_1 : \eta_1 < \eta_2$ (one-sided): Reject H_0 if u_1 is not larger than the tabulated value.

$H_0 : \eta_1 = \eta_2$ vs. $H_1 : \eta_1 \neq \eta_2$ (two-sided): Reject H_0 if $\min\{u_1, u_2\}$ is not larger than the tabulated value.

“–” indicates that H_0 is never rejected for the chosen size α and sample sizes n_1 and n_2 .

Significance level (α): 5% one-sided, 10% two-sided

n_1	n_2																								
	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
2	–	–	–	0	0	0	1	1	1	1	2	2	3	3	3	3	4	4	4	5	5	5	6	6	
3	–	0	0	1	2	2	3	4	4	5	5	6	7	7	8	9	9	10	11	11	12	13	13	14	
4	–	0	1	2	3	4	5	6	7	8	9	10	11	12	14	15	16	17	18	19	20	21	22	23	
5	0	1	2	4	5	6	8	9	11	12	13	15	16	18	19	20	22	23	25	26	28	29	30	32	
6	0	2	3	5	7	8	10	12	14	16	17	19	21	23	25	26	28	30	32	34	36	37	39	41	
7	0	2	4	6	8	11	13	15	17	19	21	24	26	28	30	33	35	37	39	41	44	46	48	50	
8	1	3	5	8	10	13	15	18	20	23	26	28	31	33	36	39	41	44	47	49	52	54	57	60	
9	1	4	6	9	12	15	18	21	24	27	30	33	36	39	42	45	48	51	54	57	60	63	66	69	
10	1	4	7	11	14	17	20	24	27	31	34	37	41	44	48	51	55	58	62	65	68	72	75	79	
11	1	5	8	12	16	19	23	27	31	34	38	42	46	50	54	57	61	65	69	73	77	81	85	89	
12	2	5	9	13	17	21	26	30	34	38	42	47	51	55	60	64	68	72	77	81	85	90	94	98	
13	2	6	10	15	19	24	28	33	37	42	47	51	56	61	65	70	75	80	84	89	94	98	103	108	
14	3	7	11	16	21	26	31	36	41	46	51	56	61	66	71	77	82	87	92	97	102	107	113	118	
15	3	7	12	18	23	28	33	39	44	50	55	61	66	72	77	83	88	94	100	105	111	116	122	128	
16	3	8	14	19	25	30	36	42	48	54	60	65	71	77	83	89	95	101	107	113	119	125	131	137	
17	3	9	15	20	26	33	39	45	51	57	64	70	77	83	89	96	102	109	115	121	128	134	141	147	
18	4	9	16	22	28	35	41	48	55	61	68	75	82	88	95	102	109	116	123	130	136	143	150	157	
19	4	10	17	23	30	37	44	51	58	65	72	80	87	94	101	109	116	123	130	138	145	152	160	167	
20	4	11	18	25	32	39	47	54	62	69	77	84	92	100	107	115	123	130	138	146	154	161	169	177	
21	5	11	19	26	34	41	49	57	65	73	81	89	97	105	113	121	130	138	146	154	162	170	179	187	
22	5	12	20	28	36	44	52	60	68	77	85	94	102	111	119	128	136	145	154	162	171	179	188	197	
23	5	13	21	29	37	46	54	63	72	81	90	98	107	116	125	134	143	152	161	170	179	189	198	207	
24	6	13	22	30	39	48	57	66	75	85	94	103	113	122	131	141	150	160	169	179	188	198	207	217	
25	6	14	23	32	41	50	60	69	79	89	98	108	118	128	137	147	157	167	177	187	197	207	217	227	

Significance level (α): 2.5% one-sided, 5% two-sided

n_1	n_2																								
	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
2	—	—	—	—	—	—	0	0	0	0	1	1	1	1	1	2	2	2	2	3	3	3	3	3	
3	—	—	—	0	1	1	2	2	3	3	4	4	5	5	6	6	7	7	8	8	9	9	10	10	
4	—	—	0	1	2	3	4	4	5	6	7	8	9	10	11	11	12	13	14	15	16	17	17	18	
5	—	0	1	2	3	5	6	7	8	9	11	12	13	14	15	17	18	19	20	22	23	24	25	27	
6	—	1	2	3	5	6	8	10	11	13	14	16	17	19	21	22	24	25	27	29	30	32	33	35	
7	—	1	3	5	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	
8	0	2	4	6	8	10	13	15	17	19	22	24	26	29	31	34	36	38	41	43	45	48	50	53	
9	0	2	4	7	10	12	15	17	20	23	26	28	31	34	37	39	42	45	48	50	53	56	59	62	
10	0	3	5	8	11	14	17	20	23	26	29	33	36	39	42	45	48	52	55	58	61	64	67	71	
11	0	3	6	9	13	16	19	23	26	30	33	37	40	44	47	51	55	58	62	65	69	73	76	80	
12	1	4	7	11	14	18	22	26	29	33	37	41	45	49	53	57	61	65	69	73	77	81	85	89	
13	1	4	8	12	16	20	24	28	33	37	41	45	50	54	59	63	67	72	76	80	85	89	94	98	
14	1	5	9	13	17	22	26	31	36	40	45	50	55	59	64	69	74	78	83	88	93	98	102	107	
15	1	5	10	14	19	24	29	34	39	44	49	54	59	64	70	75	80	85	90	96	101	106	111	117	
16	1	6	11	15	21	26	31	37	42	47	53	59	64	70	75	81	86	92	98	103	109	115	120	126	
17	2	6	11	17	22	28	34	39	45	51	57	63	69	75	81	87	93	99	105	111	117	123	129	135	
18	2	7	12	18	24	30	36	42	48	55	61	67	74	80	86	93	99	106	112	119	125	132	138	145	
19	2	7	13	19	25	32	38	45	52	58	65	72	78	85	92	99	106	113	119	126	133	140	147	154	
20	2	8	14	20	27	34	41	48	55	62	69	76	83	90	98	105	112	119	127	134	141	149	156	163	
21	3	8	15	22	29	36	43	50	58	65	73	80	88	96	103	111	119	126	134	142	150	157	165	173	
22	3	9	16	23	30	38	45	53	61	69	77	85	93	101	109	117	125	133	141	150	158	166	174	182	
23	3	9	17	24	32	40	48	56	64	73	81	89	98	106	115	123	132	140	149	157	166	175	183	192	
24	3	10	17	25	33	42	50	59	67	76	85	94	102	111	120	129	138	147	156	165	174	183	192	201	
25	3	10	18	27	35	44	53	62	71	80	89	98	107	117	126	135	145	154	163	173	182	192	201	211	

Significance level (α): 1% one-sided, 2% two-sided

n_1	n_2																								
	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
2	−	−	−	−	−	−	−	−	−	−	−	0	0	0	0	0	0	1	1	1	1	1	1	1	
3	−	−	−	−	−	0	0	1	1	1	2	2	2	3	3	4	4	4	5	5	6	6	6	7	
4	−	−	−	0	1	1	2	3	3	4	5	5	6	7	7	8	9	9	10	11	11	12	13	13	
5	−	−	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	
6	−	−	1	2	3	4	6	7	8	9	11	12	13	15	16	18	19	20	22	23	24	26	27	29	
7	−	0	1	3	4	6	7	9	11	12	14	16	17	19	21	23	24	26	28	30	31	33	35	36	
8	−	0	2	4	6	7	9	11	13	15	17	20	22	24	26	28	30	32	34	36	38	40	42	45	
9	−	1	3	5	7	9	11	14	16	18	21	23	26	28	31	33	36	38	40	43	45	48	50	53	
10	−	1	3	6	8	11	13	16	19	22	24	27	30	33	36	38	41	44	47	50	53	55	58	61	
11	−	1	4	7	9	12	15	18	22	25	28	31	34	37	41	44	47	50	53	57	60	63	66	70	
12	−	2	5	8	11	14	17	21	24	28	31	35	38	42	46	49	53	56	60	64	67	71	75	78	
13	0	2	5	9	12	16	20	23	27	31	35	39	43	47	51	55	59	63	67	71	75	79	83	87	
14	0	2	6	10	13	17	22	26	30	34	38	43	47	51	56	60	65	69	73	78	82	87	91	95	
15	0	3	7	11	15	19	24	28	33	37	42	47	51	56	61	66	70	75	80	85	90	94	99	104	
16	0	3	7	12	16	21	26	31	36	41	46	51	56	61	66	71	76	82	87	92	97	102	108	113	
17	0	4	8	13	18	23	28	33	38	44	49	55	60	66	71	77	82	88	93	99	105	110	116	122	
18	0	4	9	14	19	24	30	36	41	47	53	59	65	70	76	82	88	94	100	106	112	118	124	130	
19	1	4	9	15	20	26	32	38	44	50	56	63	69	75	82	88	94	101	107	113	120	126	133	139	
20	1	5	10	16	22	28	34	40	47	53	60	67	73	80	87	93	100	107	114	121	127	134	141	148	
21	1	5	11	17	23	30	36	43	50	57	64	71	78	85	92	99	106	113	121	128	135	142	150	157	
22	1	6	11	18	24	31	38	45	53	60	67	75	82	90	97	105	112	120	127	135	143	150	158	166	
23	1	6	12	19	26	33	40	48	55	63	71	79	87	94	102	110	118	126	134	142	150	158	167	175	
24	1	6	13	20	27	35	42	50	58	66	75	83	91	99	108	116	124	133	141	150	158	167	175	184	
25	1	7	13	21	29	36	45	53	61	70	78	87	95	104	113	122	130	139	148	157	166	175	184	192	

6 Correlation

The table below contains the critical values for testing whether the linear correlation coefficient is 0. The test statistic used is the empirical correlation coefficient r computed from a sample of size n .

The test is based on the same assumptions as the simple linear model, i.e. $Y_i | X_i = x_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$, with the Y_i being independent. This assumption for example holds when (X_i, Y_i) are i.i.d. realisations from a bivariate normal distribution.

$H_0 : \rho \leq 0$ vs. $H_1 : \rho > 0$ (one-sided): Reject H_0 if r is greater than the tabulated value.
 $H_0 : \rho \geq 0$ vs. $H_1 : \rho < 0$ (one-sided): Reject H_0 if $-r$ is greater than the tabulated value.
 $H_0 : \rho = 0$ vs. $H_1 : \rho \neq 0$ (two-sided): Reject H_0 if $|r|$ is greater than the tabulated value.

size α (one-sided)					size α (one-sided)				

Note that confidence intervals and tests for the linear correlation coefficient can also be constructed using Fisher's z -transform, which is tabulated in Section 7.

7 Fisher's z -transform

The function tabulated below is Fisher's z -transform

$$z(r) \equiv \tanh^{-1}(r) = \frac{1}{2} \log \left(\frac{1+r}{1-r} \right).$$

r	.000	.001	.002	.003	.004	.005	.006	.007	.008	.009
0.00	0.0000	0.0010	0.0020	0.0030	0.0040	0.0050	0.0060	0.0070	0.0080	0.0090
0.01	0.0100	0.0110	0.0120	0.0130	0.0140	0.0150	0.0160	0.0170	0.0180	0.0190
0.02	0.0200	0.0210	0.0220	0.0230	0.0240	0.0250	0.0260	0.0270	0.0280	0.0290
0.03	0.0300	0.0310	0.0320	0.0330	0.0340	0.0350	0.0360	0.0370	0.0380	0.0390
0.04	0.0400	0.0410	0.0420	0.0430	0.0440	0.0450	0.0460	0.0470	0.0480	0.0490
0.05	0.0500	0.0510	0.0520	0.0530	0.0541	0.0551	0.0561	0.0571	0.0581	0.0591
0.06	0.0601	0.0611	0.0621	0.0631	0.0641	0.0651	0.0661	0.0671	0.0681	0.0691
0.07	0.0701	0.0711	0.0721	0.0731	0.0741	0.0751	0.0761	0.0772	0.0782	0.0792
0.08	0.0802	0.0812	0.0822	0.0832	0.0842	0.0852	0.0862	0.0872	0.0882	0.0892
0.09	0.0902	0.0913	0.0923	0.0933	0.0943	0.0953	0.0963	0.0973	0.0983	0.0993
0.10	0.1003	0.1013	0.1024	0.1034	0.1044	0.1054	0.1064	0.1074	0.1084	0.1094
0.11	0.1104	0.1115	0.1125	0.1135	0.1145	0.1155	0.1165	0.1175	0.1186	0.1196
0.12	0.1206	0.1216	0.1226	0.1236	0.1246	0.1257	0.1267	0.1277	0.1287	0.1297
0.13	0.1307	0.1318	0.1328	0.1338	0.1348	0.1358	0.1368	0.1379	0.1389	0.1399
0.14	0.1409	0.1419	0.1430	0.1440	0.1450	0.1460	0.1471	0.1481	0.1491	0.1501
0.15	0.1511	0.1522	0.1532	0.1542	0.1552	0.1563	0.1573	0.1583	0.1593	0.1604
0.16	0.1614	0.1624	0.1634	0.1645	0.1655	0.1665	0.1676	0.1686	0.1696	0.1706
0.17	0.1717	0.1727	0.1737	0.1748	0.1758	0.1768	0.1779	0.1789	0.1799	0.1809
0.18	0.1820	0.1830	0.1841	0.1851	0.1861	0.1872	0.1882	0.1892	0.1903	0.1913
0.19	0.1923	0.1934	0.1944	0.1955	0.1965	0.1975	0.1986	0.1996	0.2007	0.2017
0.20	0.2027	0.2038	0.2048	0.2059	0.2069	0.2079	0.2090	0.2100	0.2111	0.2121
0.21	0.2132	0.2142	0.2153	0.2163	0.2174	0.2184	0.2195	0.2205	0.2216	0.2226
0.22	0.2237	0.2247	0.2258	0.2268	0.2279	0.2289	0.2300	0.2310	0.2321	0.2331
0.23	0.2342	0.2352	0.2363	0.2374	0.2384	0.2395	0.2405	0.2416	0.2427	0.2437
0.24	0.2448	0.2458	0.2469	0.2480	0.2490	0.2501	0.2512	0.2522	0.2533	0.2543
0.25	0.2554	0.2565	0.2575	0.2586	0.2597	0.2608	0.2618	0.2629	0.2640	0.2650
0.26	0.2661	0.2672	0.2683	0.2693	0.2704	0.2715	0.2726	0.2736	0.2747	0.2758
0.27	0.2769	0.2779	0.2790	0.2801	0.2812	0.2823	0.2833	0.2844	0.2855	0.2866
0.28	0.2877	0.2888	0.2899	0.2909	0.2920	0.2931	0.2942	0.2953	0.2964	0.2975
0.29	0.2986	0.2997	0.3008	0.3018	0.3029	0.3040	0.3051	0.3062	0.3073	0.3084
0.30	0.3095	0.3106	0.3117	0.3128	0.3139	0.3150	0.3161	0.3172	0.3183	0.3194
0.31	0.3205	0.3217	0.3228	0.3239	0.3250	0.3261	0.3272	0.3283	0.3294	0.3305
0.32	0.3316	0.3328	0.3339	0.3350	0.3361	0.3372	0.3383	0.3395	0.3406	0.3417
0.33	0.3428	0.3440	0.3451	0.3462	0.3473	0.3484	0.3496	0.3507	0.3518	0.3530
0.34	0.3541	0.3552	0.3564	0.3575	0.3586	0.3598	0.3609	0.3620	0.3632	0.3643
0.35	0.3654	0.3666	0.3677	0.3689	0.3700	0.3712	0.3723	0.3734	0.3746	0.3757
0.36	0.3769	0.3780	0.3792	0.3803	0.3815	0.3826	0.3838	0.3850	0.3861	0.3873
0.37	0.3884	0.3896	0.3907	0.3919	0.3931	0.3942	0.3954	0.3966	0.3977	0.3989
0.38	0.4001	0.4012	0.4024	0.4036	0.4047	0.4059	0.4071	0.4083	0.4094	0.4106
0.39	0.4118	0.4130	0.4142	0.4153	0.4165	0.4177	0.4189	0.4201	0.4213	0.4225
0.40	0.4236	0.4248	0.4260	0.4272	0.4284	0.4296	0.4308	0.4320	0.4332	0.4344
0.41	0.4356	0.4368	0.4380	0.4392	0.4404	0.4416	0.4428	0.4441	0.4453	0.4465
0.42	0.4477	0.4489	0.4501	0.4513	0.4526	0.4538	0.4550	0.4562	0.4574	0.4587
0.43	0.4599	0.4611	0.4624	0.4636	0.4648	0.4660	0.4673	0.4685	0.4698	0.4710
0.44	0.4722	0.4735	0.4747	0.4760	0.4772	0.4784	0.4797	0.4809	0.4822	0.4834
0.45	0.4847	0.4860	0.4872	0.4885	0.4897	0.4910	0.4922	0.4935	0.4948	0.4960
0.46	0.4973	0.4986	0.4999	0.5011	0.5024	0.5037	0.5049	0.5062	0.5075	0.5088
0.47	0.5101	0.5114	0.5126	0.5139	0.5152	0.5165	0.5178	0.5191	0.5204	0.5217
0.48	0.5230	0.5243	0.5256	0.5269	0.5282	0.5295	0.5308	0.5321	0.5334	0.5347
0.49	0.5361	0.5374	0.5387	0.5400	0.5413	0.5427	0.5440	0.5453	0.5466	0.5480

→

r	r									
	.000	.001	.002	.003	.004	.005	.006	.007	.008	.009
0.50	0.5493	0.5506	0.5520	0.5533	0.5547	0.5560	0.5573	0.5587	0.5600	0.5614
0.51	0.5627	0.5641	0.5654	0.5668	0.5682	0.5695	0.5709	0.5722	0.5736	0.5750
0.52	0.5763	0.5777	0.5791	0.5805	0.5818	0.5832	0.5846	0.5860	0.5874	0.5888
0.53	0.5901	0.5915	0.5929	0.5943	0.5957	0.5971	0.5985	0.5999	0.6013	0.6027
0.54	0.6042	0.6056	0.6070	0.6084	0.6098	0.6112	0.6127	0.6141	0.6155	0.6169
0.55	0.6184	0.6198	0.6213	0.6227	0.6241	0.6256	0.6270	0.6285	0.6299	0.6314
0.56	0.6328	0.6343	0.6358	0.6372	0.6387	0.6401	0.6416	0.6431	0.6446	0.6460
0.57	0.6475	0.6490	0.6505	0.6520	0.6535	0.6550	0.6565	0.6580	0.6595	0.6610
0.58	0.6625	0.6640	0.6655	0.6670	0.6685	0.6700	0.6716	0.6731	0.6746	0.6761
0.59	0.6777	0.6792	0.6807	0.6823	0.6838	0.6854	0.6869	0.6885	0.6900	0.6916
0.60	0.6931	0.6947	0.6963	0.6978	0.6994	0.7010	0.7026	0.7042	0.7057	0.7073
0.61	0.7089	0.7105	0.7121	0.7137	0.7153	0.7169	0.7185	0.7201	0.7218	0.7234
0.62	0.7250	0.7266	0.7283	0.7299	0.7315	0.7332	0.7348	0.7365	0.7381	0.7398
0.63	0.7414	0.7431	0.7447	0.7464	0.7481	0.7498	0.7514	0.7531	0.7548	0.7565
0.64	0.7582	0.7599	0.7616	0.7633	0.7650	0.7667	0.7684	0.7701	0.7718	0.7736
0.65	0.7753	0.7770	0.7788	0.7805	0.7823	0.7840	0.7858	0.7875	0.7893	0.7910
0.66	0.7928	0.7946	0.7964	0.7981	0.7999	0.8017	0.8035	0.8053	0.8071	0.8089
0.67	0.8107	0.8126	0.8144	0.8162	0.8180	0.8199	0.8217	0.8236	0.8254	0.8273
0.68	0.8291	0.8310	0.8328	0.8347	0.8366	0.8385	0.8404	0.8423	0.8441	0.8460
0.69	0.8480	0.8499	0.8518	0.8537	0.8556	0.8576	0.8595	0.8614	0.8634	0.8653
0.70	0.8673	0.8693	0.8712	0.8732	0.8752	0.8772	0.8792	0.8812	0.8832	0.8852
0.71	0.8872	0.8892	0.8912	0.8933	0.8953	0.8973	0.8994	0.9014	0.9035	0.9056
0.72	0.9076	0.9097	0.9118	0.9139	0.9160	0.9181	0.9202	0.9223	0.9245	0.9266
0.73	0.9287	0.9309	0.9330	0.9352	0.9373	0.9395	0.9417	0.9439	0.9461	0.9483
0.74	0.9505	0.9527	0.9549	0.9571	0.9594	0.9616	0.9639	0.9661	0.9684	0.9707
0.75	0.9730	0.9752	0.9775	0.9798	0.9822	0.9845	0.9868	0.9892	0.9915	0.9939
0.76	0.9962	0.9986	1.0010	1.0034	1.0058	1.0082	1.0106	1.0130	1.0154	1.0179
0.77	1.0203	1.0228	1.0253	1.0277	1.0302	1.0327	1.0352	1.0378	1.0403	1.0428
0.78	1.0454	1.0479	1.0505	1.0531	1.0557	1.0583	1.0609	1.0635	1.0661	1.0688
0.79	1.0714	1.0741	1.0768	1.0795	1.0822	1.0849	1.0876	1.0903	1.0931	1.0958
0.80	1.0986	1.1014	1.1042	1.1070	1.1098	1.1127	1.1155	1.1184	1.1212	1.1241
0.81	1.1270	1.1299	1.1329	1.1358	1.1388	1.1417	1.1447	1.1477	1.1507	1.1538
0.82	1.1568	1.1599	1.1630	1.1660	1.1692	1.1723	1.1754	1.1786	1.1817	1.1849
0.83	1.1881	1.1914	1.1946	1.1979	1.2011	1.2044	1.2077	1.2111	1.2144	1.2178
0.84	1.2212	1.2246	1.2280	1.2315	1.2349	1.2384	1.2419	1.2454	1.2490	1.2526
0.85	1.2562	1.2598	1.2634	1.2671	1.2707	1.2745	1.2782	1.2819	1.2857	1.2895
0.86	1.2933	1.2972	1.3011	1.3050	1.3089	1.3129	1.3169	1.3209	1.3249	1.3290
0.87	1.3331	1.3372	1.3414	1.3456	1.3498	1.3540	1.3583	1.3626	1.3670	1.3714
0.88	1.3758	1.3802	1.3847	1.3892	1.3938	1.3984	1.4030	1.4077	1.4124	1.4171
0.89	1.4219	1.4268	1.4316	1.4365	1.4415	1.4465	1.4516	1.4566	1.4618	1.4670
0.90	1.4722	1.4775	1.4828	1.4882	1.4937	1.4992	1.5047	1.5103	1.5160	1.5217
0.91	1.5275	1.5334	1.5393	1.5453	1.5513	1.5574	1.5636	1.5698	1.5762	1.5826
0.92	1.5890	1.5956	1.6022	1.6089	1.6157	1.6226	1.6296	1.6366	1.6438	1.6510
0.93	1.6584	1.6658	1.6734	1.6811	1.6888	1.6967	1.7047	1.7129	1.7211	1.7295
0.94	1.7380	1.7467	1.7555	1.7645	1.7736	1.7828	1.7923	1.8019	1.8117	1.8216
0.95	1.8318	1.8421	1.8527	1.8635	1.8745	1.8857	1.8972	1.9090	1.9210	1.9333
0.96	1.9459	1.9588	1.9721	1.9857	1.9996	2.0139	2.0287	2.0439	2.0595	2.0756
0.97	2.0923	2.1095	2.1273	2.1457	2.1649	2.1847	2.2054	2.2269	2.2494	2.2729
0.98	2.2976	2.3235	2.3507	2.3796	2.4101	2.4427	2.4774	2.5147	2.5550	2.5987
0.99	2.6467	2.6996	2.7587	2.8257	2.9031	2.9945	3.1063	3.2504	3.4534	3.8002